

Cohort workbook

Applied Gen AI Research: RAG & Agent Development

This workbook accompanies the live sessions. Each block lists what to have ready before the session and what to complete after it. The starter repository and sample corpus are linked from your confirmation page.

Before session 1

- Python 3.11 or newer, with a virtual environment you can install into.
- An API key for at least one model provider.
- Clone the starter repository and run the smoke test - it should print a model response.
- Skim Lewis et al. (2020). Do not study it; we read it together.

After session 1 - Index something you care about

- Point the indexer at a corpus of your own: your docs, a wiki export, a set of PDFs.
- Write ten questions a colleague might genuinely ask of it.
- Run them. Bring the three worst answers to session 2, with the retrieved passages.

After session 2 - Rebuild the pipeline

- Swap the chunking strategy and measure the difference rather than guessing at it.
- Add sparse retrieval alongside dense; compare on your own ten questions.
- Add a reranker. Record the latency cost as well as the quality change.

After session 3 - Build the harness

- Turn your ten questions into a scored test set with known-correct passages.
- Add at least three questions the corpus cannot answer.
- Get the harness running from one command. You will use it for the rest of the cohort.

After session 4 - Wrap it in an agent

- Expose your retrieval pipeline as a tool with a clear schema.
- Implement the loop with an explicit step budget and a termination condition.
- Log every tool call and its result. Bring a trace where the agent went wrong.

After session 5 - Harden it

- Add memory. Justify to yourself what you persist and what you recompute.
- Add graceful degradation for a failing tool.
- Run the harness once more and write down what improved and what regressed.

Office hours

Bring code, a trace, or a specific question. 'It is not working' is hard to help with; a failing query with its retrieved passages takes about four minutes to diagnose.

Gen AI Live - live, small-group generative AI training. Sessions run on Google Meet or Zoom; booking, intake and material live at the website. Questions: hello@genailive.in